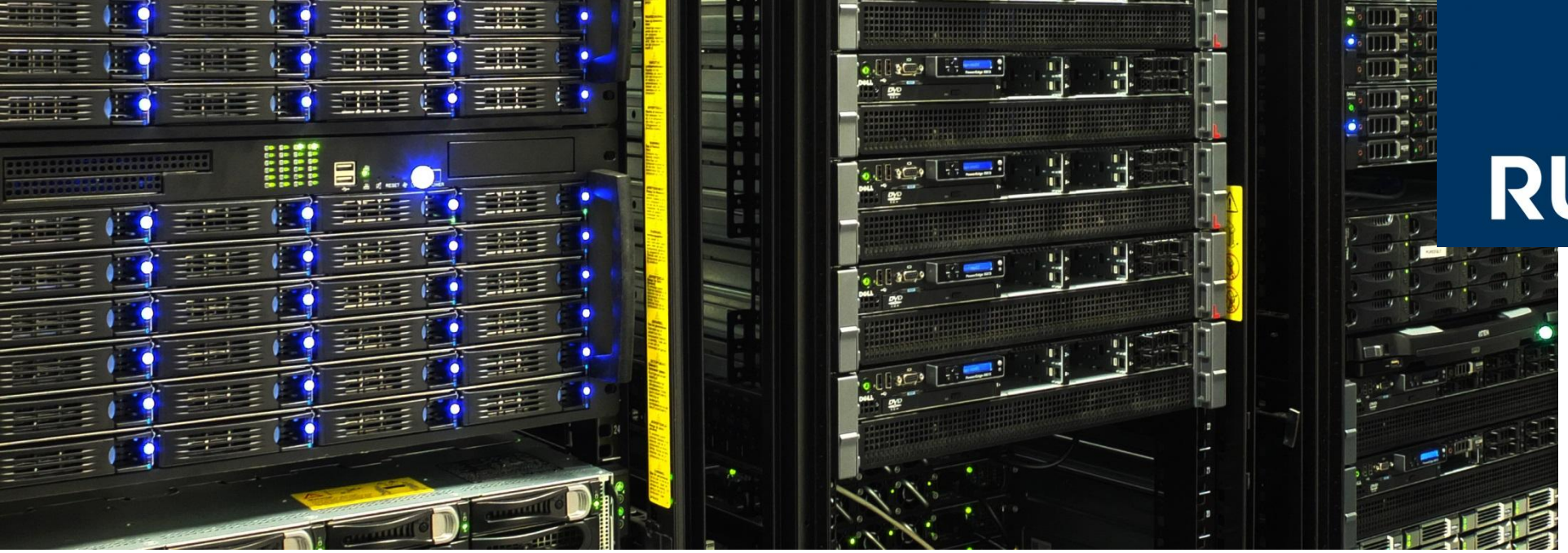


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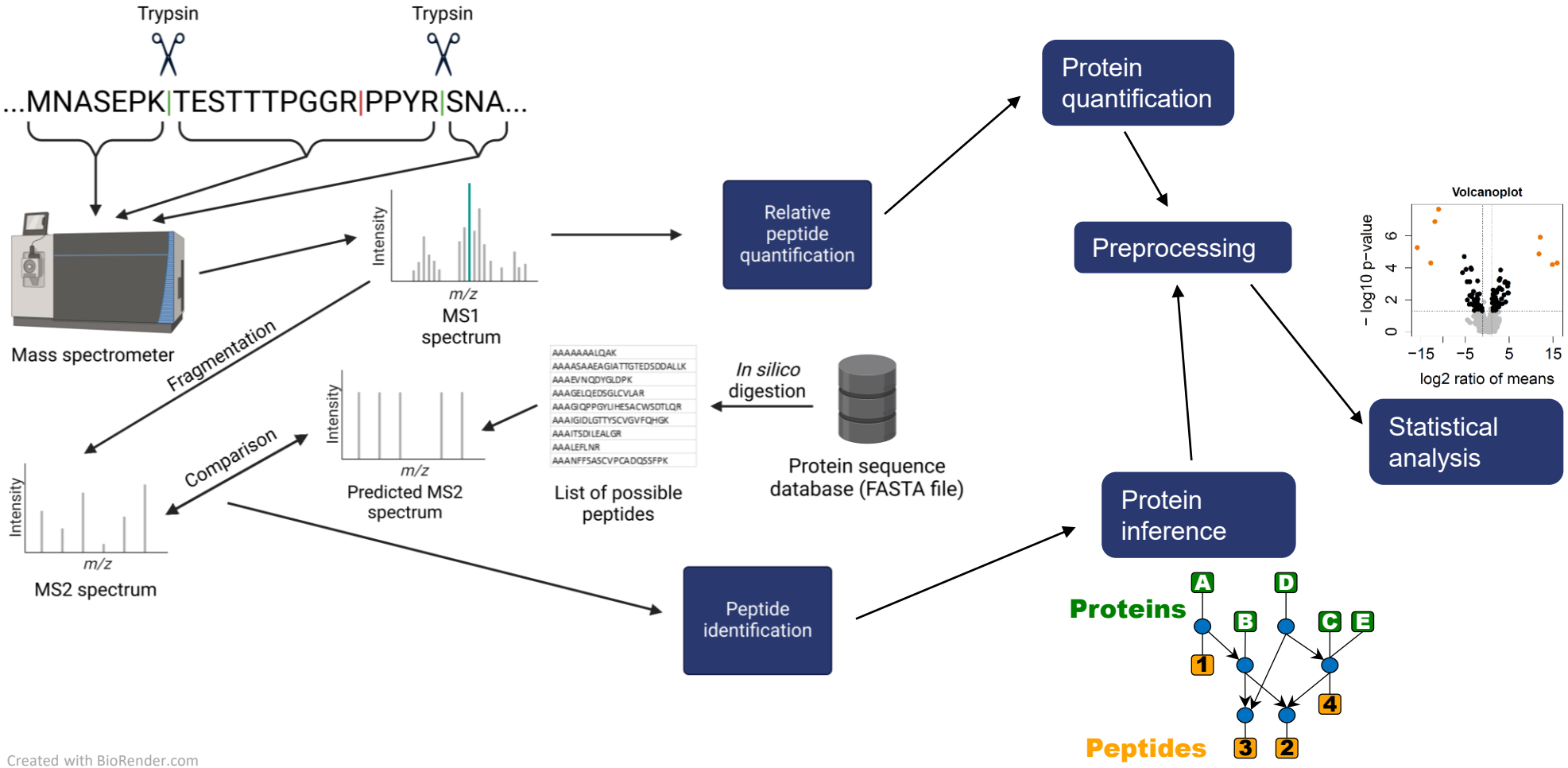
Bioinformatics for Biochemists

Tutorial Proteomics part III

Dr. Karin Schork
karin.schork@rub.de

Medizinische Fakultät
Medizinisches Proteom-Center
Medizinische Bioinformatik

Bottom-Up Proteomics Workflow



Contents

- limma and Volcano Plot
- Clustering
- Heatmaps using the ComplexHeatmap package

limma
challenge of the ordinary t-test

$$t = \frac{d}{s \cdot \sqrt{\frac{2}{n}}}$$

- Some studies have very low sample sizes (e.g., 3 vs. 3)
- Low sample sizes lead to a low power
- Additionally, the estimation of s unstable (the estimation of the variance has a high variance)
  - Underestimation:
    - One protein just happened to have very low variance by chance, not because it's truly stable, but because you got lucky with the 3 measurements
    - -> huge t-statistic and a false positive.
  - Overestimation:
    - You have a protein that is truly differentially expressed. It consistently goes up in the disease condition. But by bad luck, one of your 3 replicates is an outlier, that made it look inconsistent.
    - -> t-statistic shrinks and false negative

limma
the main idea

- Limma (Linear Models for Microarray Data) is an R package for analysis of omics data
- It incorporates the so-called „moderated t-test“
- Like the t-test it compared the intensities between 2 groups
- Estimation of s is different: It borrows information from other proteins to estimate the variance more accurately.
- Employs an Empirical Bayes framework for this.
- Can be extended to more than 2 groups or other more complex experimental designs by using linear models
- Confounding variables like age, sex etc may be added to the model

Smyth, G. K. (2004). *Linear models and empirical Bayes methods for assessing differential expression in microarray experiments.* Statistical Applications in Genetics and Molecular Biology, 3(1), Article 3

Ritchie, M.E., Phipson, B., Wu, D., Hu, Y., Law, C.W., Shi, W., and Smyth, G.K. (2015).
limma powers differential expression analyses for RNA-sequencing and microarray studies. Nucleic Acids Research 43(7), e47.

limma
Moderated t-test

Same test statistic as the ordinary t-test, but the variance is estimated in a different way:
As a combination of the overall protein variance s_0^2 and the variance of the specific protein:

$$\tilde{t}_p = \frac{d_p}{\tilde{s}_p \cdot \sqrt{\frac{2}{n}}} \qquad \tilde{s}_p^2 = \frac{d_0 s_0^2 + (2n-2) s_p^2}{d_0 + 2n - 2} \qquad \tilde{t}_p \sim t_{2n-2+d_0}$$

If n is high, you „trust“ the data more to accurately estimate s_p and it is close to the ordinary t-test.

Exemplary scenarios:

Scenario	s_p^2	s_0^2	$\tilde{s}_p^2$	Effect
Gene with suspiciously low variance	0.01	0.5	pulled **up** toward 0.5	fewer false positives
Gene with high variance	2.0	0.5	pulled **down** toward 0.5	higher power
Gene near average	0.5	0.5	stays ~0.5	no change

Limma
Example usage

```
y <- …. (matrix with protein intensities)
Group <- … (vector with grouping information)
design <- model.matrix(~ Group)
fit <- lmFit(y,design)
fit <- eBayes(fit)
topTable(fit,coef=2, number = Inf, sort.by = ``p``)
```

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$$

coef = 2 -> show results for second coefficient (group)

```
> topTable(fit,coef=2)
        logFC      AveExpr         t      P.Value   adj.P.Val         B
1   1.0261201  0.613239029  5.133348 1.067577e-05 0.001067577  3.255898
65 -0.4142890  0.132351578 -2.023881 5.065654e-02 0.984178668 -4.656835
86 -0.3849375 -0.062785795 -1.993078 5.408190e-02 0.984178668 -4.712776
6  -0.3726952 -0.030427277 -1.854613 7.207864e-02 0.984178668 -4.955723
48 -0.3678583  0.124175287 -1.797670 8.084747e-02 0.984178668 -5.051492
72  0.3688488 -0.023410981  1.727803 9.282497e-02 0.984178668 -5.165605
31 -0.3376838 -0.142492004 -1.665522 1.047219e-01 0.984178668 -5.264117
3   0.3432493  0.006803296  1.651775 1.075114e-01 0.984178668 -5.285447
90 -0.3070586 -0.256501945 -1.566593 1.261933e-01 0.984178668 -5.414233
41 -0.3205739 -0.082905186 -1.563103 1.270119e-01 0.984178668 -5.419385
```

- t: moderated t-statistic

Multiple testing problem

- t-test or limma-test on several hundreds or thousands of proteins leads to many false positives (due to type I error (α))
- Example: 1010 proteins (10 truely differentially expressed, 1000 not)
- Good power → 10 differentially expressed proteins are significant
- But you also expect about 50 (5% of 1000) false positive findings by chance
- → from ≈ 60 significant results only 10 are true

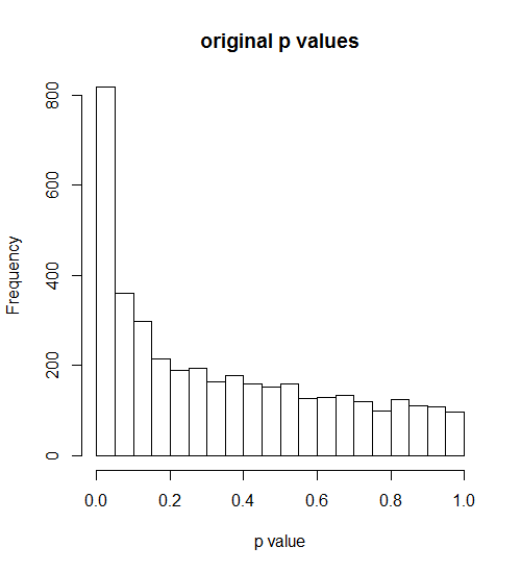
Truth	Test decision	
	do not reject H_0	reject H_0
H_0 true	☺	☹ type I error (α)
H_0 false (H_1 true)	☹ type II error (β)	☺ power (1-β)

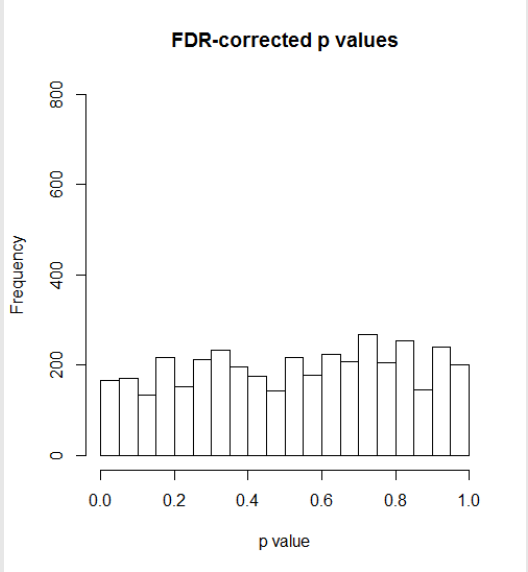
FDR Correction (Benjamini-Hochberg procedure)

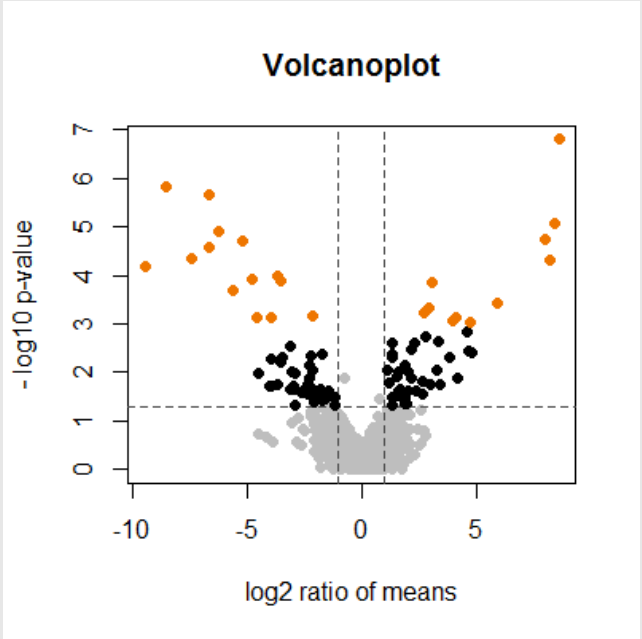
- False positives: proteins that are significant, but there is no true difference in abundance between the two groups
- FDR = false discovery rate = $\frac{\#\ false\ positives}{\#\ all\ significant\ proteins}$
- Control FDR (≤ 5%) by adjusting p-values
- p-values are enlarged

→ less proteins are significant, i.e. have a p-value below 0.05

→ remaining significant proteins are more reliable

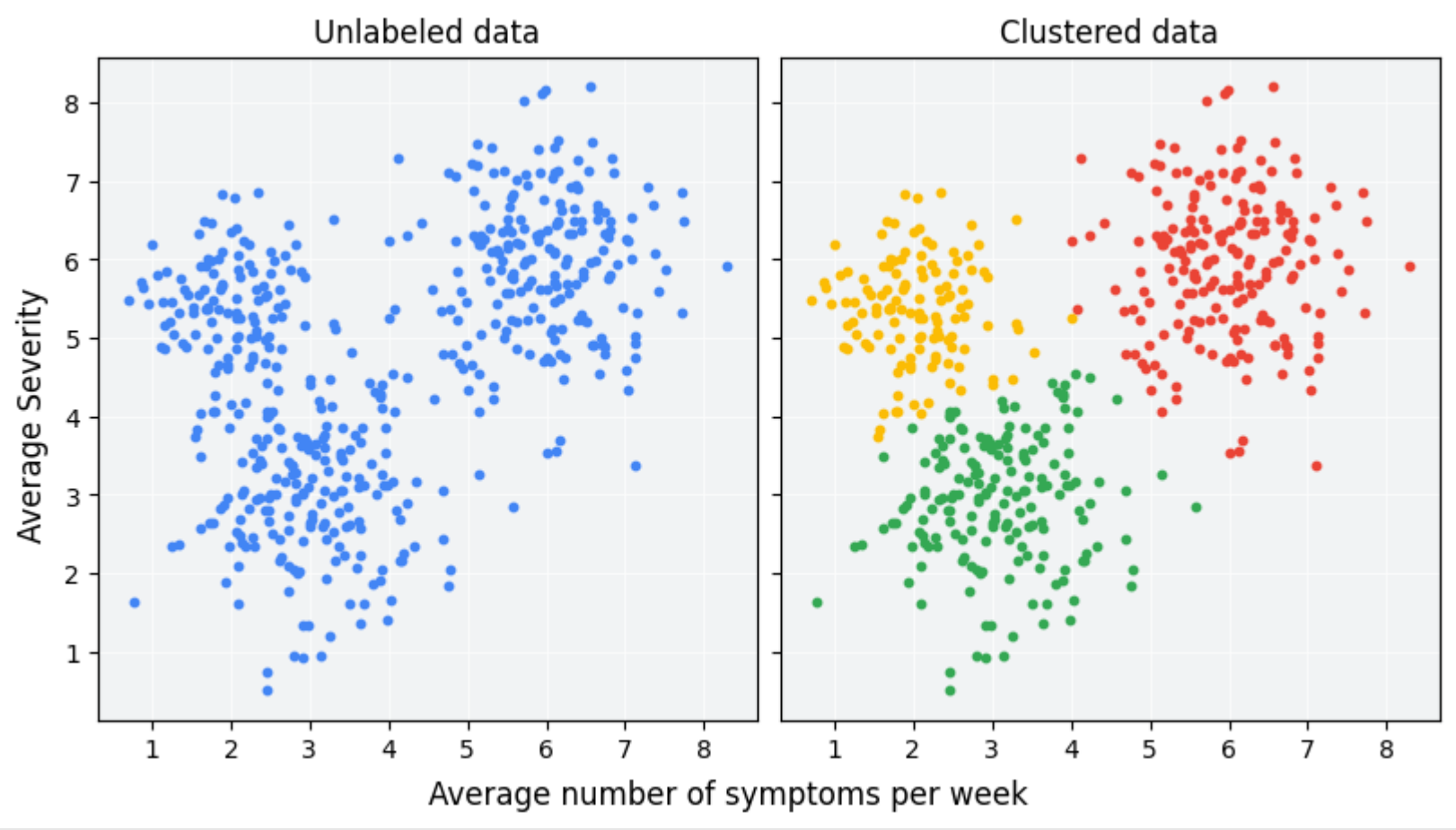






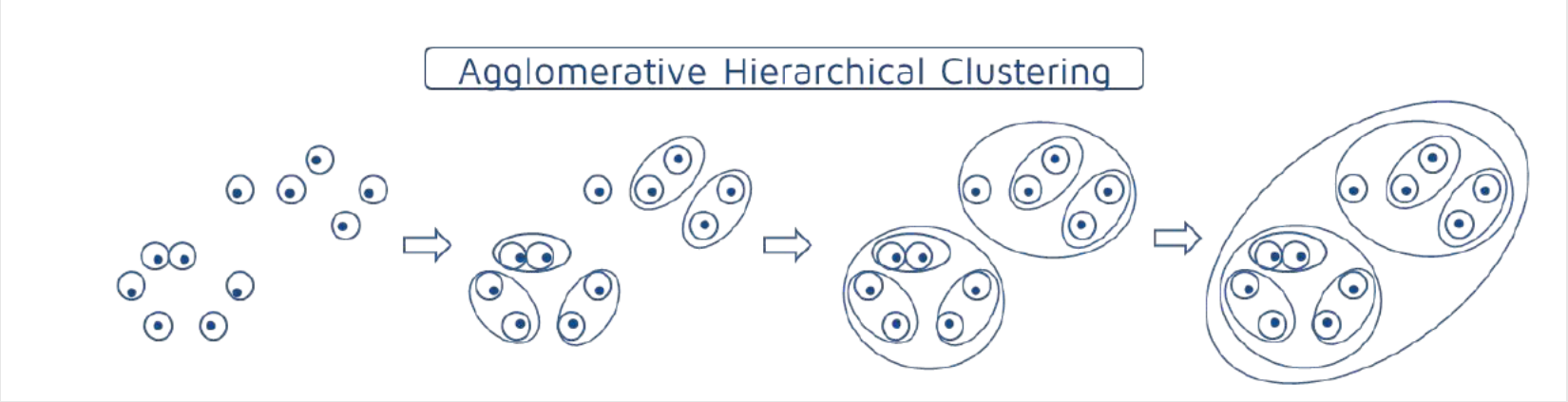
Clustering

Find groups of objects that are similar to each other



https://developers.google.com/machine-learning/clustering/overview

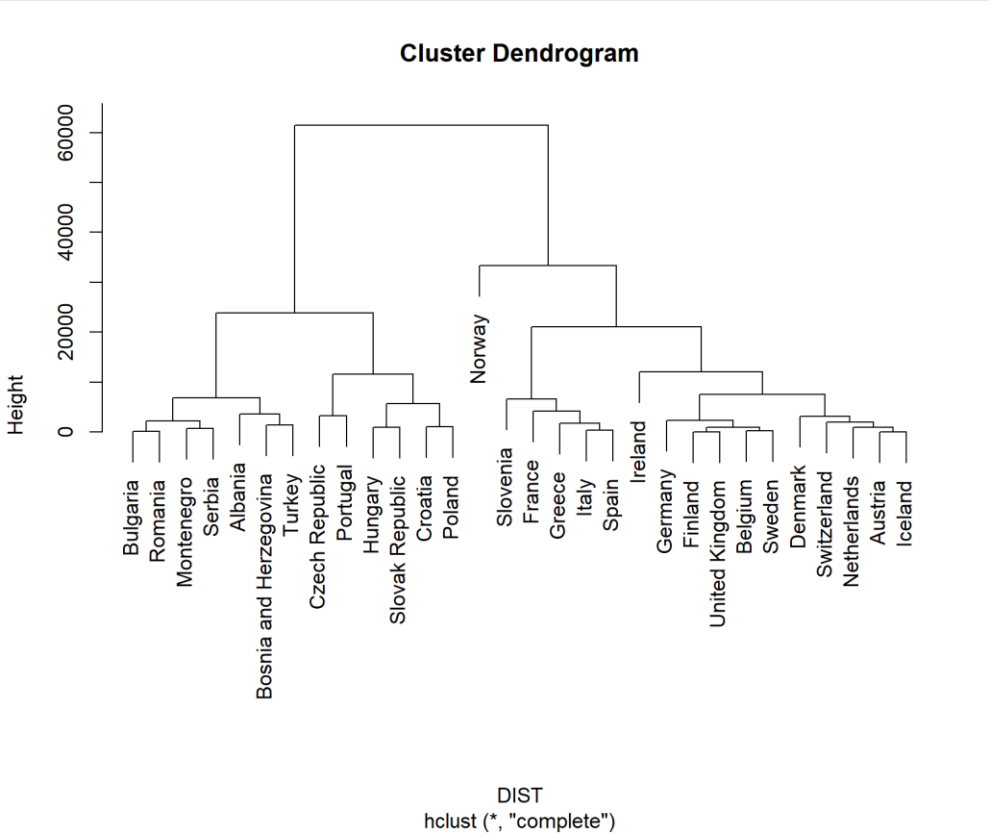
Agglomerative hierarchical clustering



1. Each data point starts as its own cluster.
2. The distance between all pairs of clusters is computed.
3. Then the two closest clusters are merged.
4. Steps 2. - 3. are repeated until all clusters are merged into a single cluster.

Dendrogram

- Typical visualization for a hierarchical clustering
- Height depends on the distance between clusters
- Dendrogram can be cut at a specific height to get a specific number of clusters



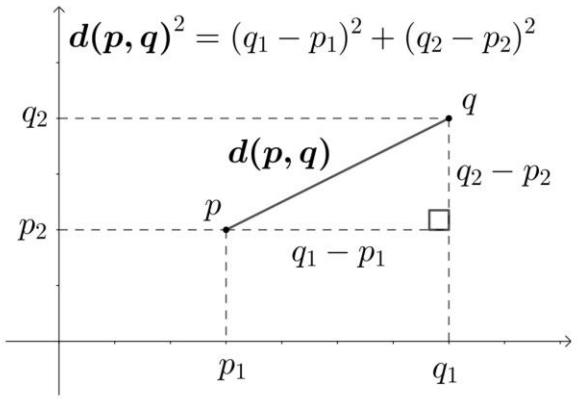
Clustering of European countries (gapminder dataset, life expectancy and GDP per capita)

Distance measures

How to calculate distance between two data points?

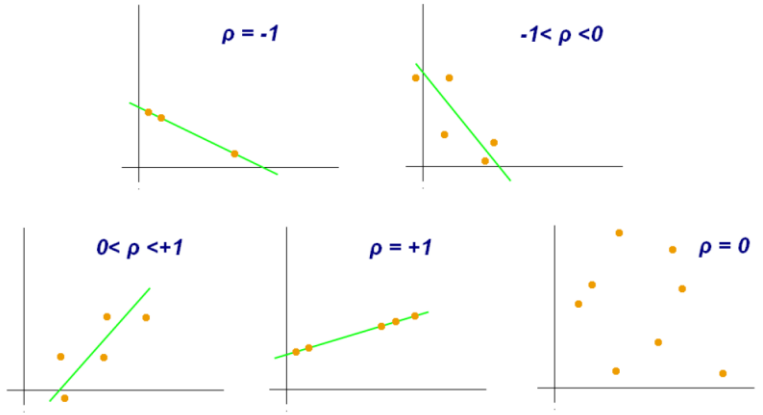
Hierarchical clustering: Euclidean distance

- The distance from classical geometry in school.
- $d(p, q) := \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$



Hierarchical clustering: Correlation-based distance

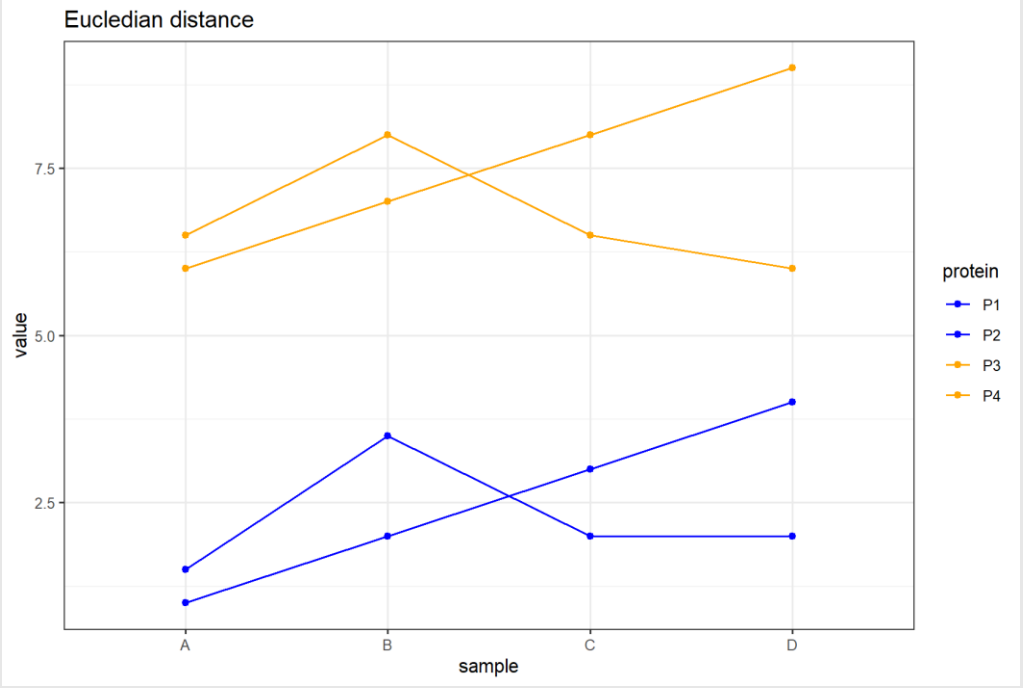
- Using correlation as distance measure.
- $d(a, b) := (1 - cor(a, b))/2$, where $cor(a, b)$ is Pearson's correlation coefficient between vectors a and b.

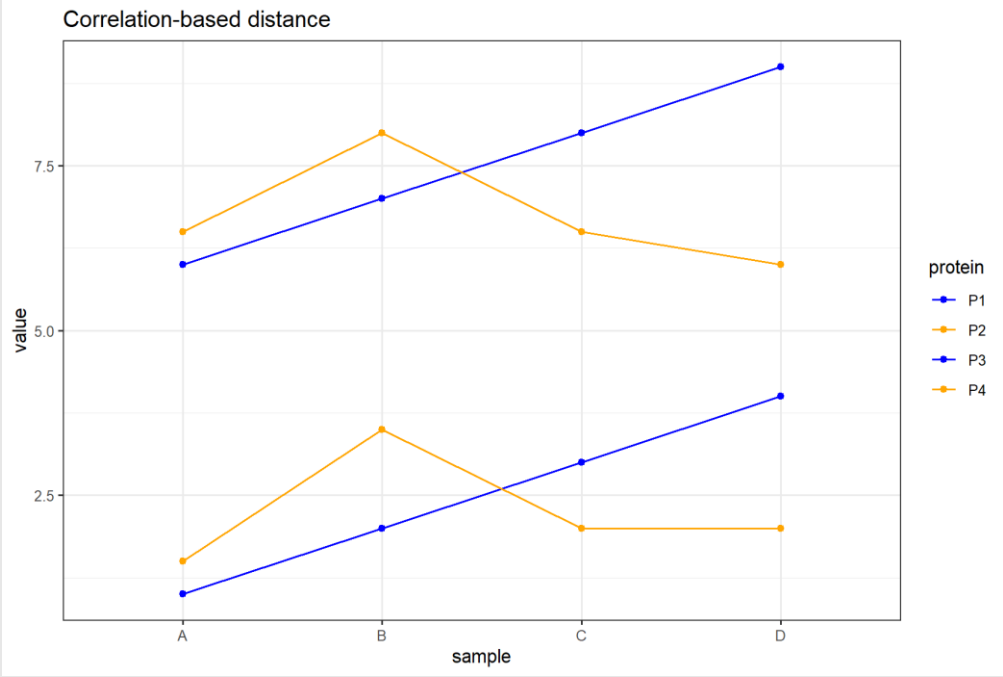


Distance measures

Euclidean distance: proteins with similar intensities are clustered together
Correlation-based distance (Pearson correlation): proteins with similar pattern are clustered together

-> correlation-based distance is preferred for proteomics, as protein intensities cannot be directly compared between different proteins

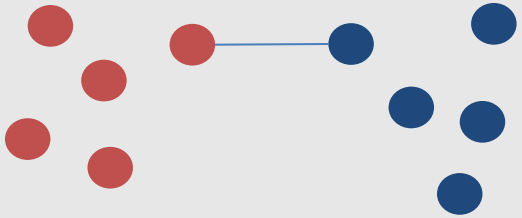




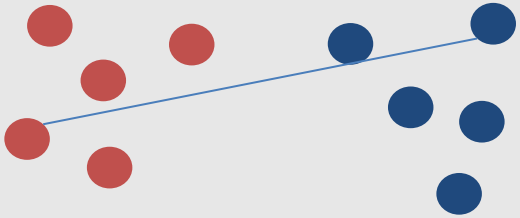
Linkage methods

Linkage: Method how to summarize distances between data points to distances between clusters

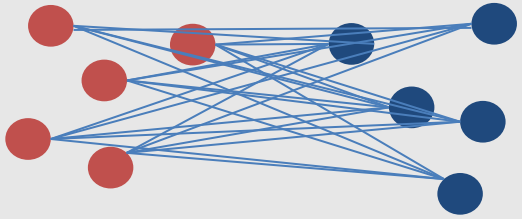
Single linkage: minimal distance between clusters



Complete linkage: maximal distance between clusters



Average linkage: average pairwise distance



For proteomics, complete linkage is often preferred, as it means that all data points have at most the maximal distance between each other

Heatmaps

- Used to get an overview over a proteomic data set (as a whole or only significant proteins)
- To get a better understanding of the structure, columns and rows are clustered and dendograms are shown
- Intensities are z-scored (substract mean, divide by SD) by protein, to show patterns
- ComplexHeatmap R package

